Traditional vs Sparsity promoting regularization

In this experience you will see the effect of regularization in image reconstruction. We will apply that mostly to MR. By the end, it should be clear to you what's the most you can achieve with traditional regularization, with sparsity-promoting ones and what are practical implications when you need to do denoising and when you instead undersample.

Task 1: Implement undersampling and see what happens ("Assign 6" notebook)

In MR you acquire the Fourier transform of the image (the "k-space"). The Nyquist theorem says that, basically, you need $N_x \times N_y$ points in k-space to reconstruct an $N_x \times N_y$ pixels image. What happens if you skip some points in the k-space?

If you don't sample a point in the k-space, it means it's unknown. Your best bet to do an analytical reconstruction is doing the inverse of the acquired data, setting to zero the points not acquired. Try this! This will result of course in some "noise".

Remember that in a real-world "standard" MR you acquire a whole line of the k-space at a time. So it would be more common to skip whole lines rather than single points.

Anyway, try the following strategies and compare the resulting images qualitatively and by comparing the MSE with the true image (read the implementation notes later)

- Skip 3 lines out of 4 in the k-space, regularly
- Skip every 3 points out of 4 in the k-space
- Skip randomly 75% of the points in the k-space, uniform probability
- Skip randomly 75% of the lines in the k-space, uniform probability
- Skip randomly 75% of the points in the k-space, with decreasing probability (to maintain) with increasing frequency (e.g. $p(f) = e^{-\alpha f}$, with f the "radial" frequency)
- Skip randomly 75% of the lines in the k-space, with decreasing probability (to maintain) with increasing frequency (e.g. $p(f) = e^{-\alpha}$

Implementation notes:

- The $(0,0)$ frequency has the integral of the image. This can never be skipped.
- The very low frequencies contain information about the overall structure of the image, you might want to keep the very first elements (e.g.: [:10,:10]) to avoid gross defects
- The Fourier transform of a real-valued array is symmetric around the 0 frequency. E.g.: kspace[-Nx/2:,:] = np.conj(kspace[:Nx/2,:]). Therefore if you keep one element out of 4, you're actually keeping one element out of 2!
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Task 2: Implements the traditional regularized reconstruction (same notebook)

The most trivial regularized reconstruction is $\|Ax - y\|^2 + \beta\|\nabla x\|^2$. The regularization used penalizes solutions with too-strong pixel-to-pixel variations. Note that for undersampled MR $A = M\mathcal{F}$, with M the binary mask of the k-space points not acquired. Implement it and judge, qualitatively and using MSE, what happens when you try to do the regularization. Also, monitor the data fidelity term, what can you say about it?

Task 3: Implement ISTA using the wavelet transform for undersampled MR (Assign 7)

First, do the introduction to sparse reconstruction notebook.

Wavelets:

A wavelet transform is just a multiscale version of the Fourier transform: instead of summing image values weighting by sinusoids, you analyze the image with small, localized filters at different scales. It's empirically seen that wavelet coefficients are sparse for natural images. Use the pywt library, pywt.wavedec2 gives you those coefficients while pywt.waverec2 puts the image back together. Conveniently, for wavelets $W^T = W^{-1}$ (just as in Fourier, from which they're inspired)

Notebook:

Implement ISTA for this problem. Use as your assumption that the solution is sparse in the wavelet transform basis. The optimization becomes $\|M\mathcal{F}W^{-1}w - y\|^2 + \beta|w|_1$, where W is the wavelet transform. Here, your image is $x = W^{-1}w$.

Compare the MSE and the image quality in the different subsampling schemes. Compare, at identical subsampling, the best MSE you can achieve with sparsity promoting regularization vs the traditional one (previous notebook).

When you've done this, go working in the next regime: fully (or mostly) sampled k-space but in presence of noise. Compare here how well the previous and the sparse-promoting regularization work.

Results analysis

Reflect about the following points:

- Is it equally easy to reconstruct images with random sub-sampling compared to images with a subsampling pattern? If not, why?
- Is $\|\nabla x\|^2$ help in any way when dealing with undersampled images?
- What's (in both experiments!) the data fidelity loss value you achieve? How does it compare to the one you achieve in an acquisition with noise (and comparable final MSE)?

- If you want, count the number of non-zero elements in w. Is optimizing $|w|$ effective in promoting sparsity (i.e.: most coefficients go should be zero)
- Compare the β values needed to achieve an effective regularization in presence of noise to that of undersampled images. Are they different? If so, why?
- For the denoising case, compare the MSE you achieve (over different β for the same data) when using L_1 (sparse wavelets) vs L_2 ($\|\nabla x\|^2$). Which works better in general?

Bonus task (natural image restoration):

Take a picture. A natural image shot with your phone. Try doing “compressive sensing” here (image restoration) using wavelet sparsity. Try 2 different cases

- Mask out 75% of the original pixels, chosen randomly (the theoretically easiest setup)
- Mask out a region manually somewhere.

BTW, try that with 2 different pictures: an image of something which is very regular (e.g. a building façade) and an image of something less ordered (e.g.: a landscape)